

Review

Frugal approach to innovation: State of the art and future perspectives

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ABSTRACT

Frugal innovation has recently become a relevant topic in the public domain. The real challenge for frugal innovation is to introduce something new or different (innovating) whilst being sparing with resources (frugal). Traditionally, it has been investigated in relation to emerging and developing countries. However, scholars and institutions have pointed out that the phenomenon is growing in relevance in advanced economies too. As a consequence, some scholars have started soliciting attempts to systematize this emerging field of research. Based on these premises, the study aims to: explore how the frugal innovation concept has developed over the years; analyze the main key findings from the extant literature on frugal innovation to lay a foundation for subsequent works, by identifying gaps in current knowledge; highlight new directions for future research. In doing so, we carried out a systematic literature review (SLR) and analyzed 113 articles that have been published since 1990. We asked experts to validate the main research gaps we had identified through the SLR. As a whole, results show that experts identify - for both emerging and developing countries and for advanced economies - various relevant future research directions.

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1. Introduction

Frugal innovation has recently become a relevant topic in the public domain, soliciting further study, especially since several frugal innovation-evangelists including entrepreneurs, managers, and business consultants have begun to exploit its business applications and disruptive potentials (Rao, 2013). According to the dictionary, the word “frugal” means “economical in use, or spending”, “requiring little expense or few resources”, but also “living without waste”.

The real challenge for frugal innovation is to introduce *something new or different* (innovating) whilst being *sparing* with resources (frugal). The concept of “frugality” in innovation, first introduced in emerging markets, was pushed by the necessity of satisfying the needs of Bottom of the Pyramid (BoP) consumers with market-specific solutions characterized by high added-value and low costs (Radjou et al., 2012; Prahalad, 2005). In this respect, Hart and Christensen (2002) were among the first to stress the importance of “Disrupting the Pyramid” by linking the concept of disruptive innovation to emerging economies. According to Rao (2013), frugal innovation represents a disruptive force that is able to make mainstream companies adopt frugality and capture large market shares in emerging countries. Frugal innovation is therefore increasingly attracting the attention of scholars and practitioners due to its potential for under-exploited markets, i.e. those on a low income – including Western countries.

Traditionally, frugal approaches have been analyzed and implemented in emerging or developing countries (Radjou et al., 2012). However, scholars have pointed out that the phenomenon is growing in importance in advanced economies too (Tiwarei et al., 2016; EU EU - European Commission, 2016; Nesta, 2016). Recently, the European Union (2014), in providing policy directions for dealing with the “Grand Challenges” in the global context, has underlined that frugal innovation may be applied in advanced economies and in specific sectors such as those linked to the Key Enabling Technologies (KETs). In this regard, India's efforts to develop frugal innovation have been highlighted as an area of potential learning for Europe.

Scholars have identified, within industrialized countries, specific customer groups that are considered “over-served”, i.e. unsatisfied non-consumers, as well as BoP consumers in advanced countries (Angot and Plé, 2015; Matzler et al., 2014). In order to meet the expectations of such consumers, high-tech products could be adapted or eventually re-engineered to incorporate basic minimum features (Matzler et al., 2014). Within this context, companies can choose to adapt their strategies to the principle of basic functionality at low cost.

Recently, many contributions have been published on the topic of frugal innovation and on similar concepts/definitions that come under the umbrella of resource-constrained innovations. Some scholars have carried out research based on systematic literature review (SLR) and they have proposed different types of systematization on the basis of the focus of analysis (Agarwal et al., 2017; Brem and Wolfram, 2014; Hossain, 2016; Weyrauch and Herstatt, 2016). Some studies, by focusing on the systematization of various terminologies, have considered the differences among the various typologies of innovations proposed by scholars over the years. Specifically, Brem and Wolfram (2014) analyzed the differences among the various types of innovation (jugaad, frugal innovation, frugal engineering, constraint-based innovation, Gandhian innovation, catalytic innovation, grassroots innovation, indigenous innovation, and reverse innovation) by identifying categories that are useful to point out similarities and differences. Conversely, the study developed by Weyrauch and Herstatt (2016) aimed at clarifying what distinguishes frugal innovation from other types of

innovation, by identifying three main criteria of classification. Other studies have tried to organize and summarize research on frugal and resource constraint-based innovation. In particular, Hossain (2016) considered sixty-two contributions (including articles, books, book chapters, and working papers) within the field of frugal innovation, and provided some descriptive statistics for the selected articles (by country, journal, keyword, and investigated sector). Then, the author selected 117 cases of frugal innovation and analyzed them by sectors, so as to provide a description of the evolution of frugal innovation in different industries. More recently, Agarwal et al. (2017) have developed a systematic literature review of constraint-based innovations based on 117 contributions (including articles, books, and conference proceedings). The study provides a descriptive quantitative analysis of the articles (by keyword, geographic focus, and industry) as well as a quantitative analysis – a hierarchical cluster analysis – that is useful to identify the fundamental dimensions of constraint-based innovations.

Despite these useful attempts, there is still a need for a qualitative analysis that investigates the key findings of the literature on frugal innovation, as well as a comprehensive discussion on the most promising research streams.

Against this backdrop, this study aims to contribute to the existing literature by answering the following research questions:

- 1) How has the concept of frugal innovation developed over the years?
- 2) What is the focus of current research on frugal innovation, and which are the key findings of the literature published so far?
- 3) What are the most relevant lines of future research according to the experts in the field?

Therefore, our study differs from previous works in that it provides a qualitative analysis of the main insights arising from the literature, and by identifying the most promising future research directions according to the experts in the field.

To reach these objectives, a systematic quantitative and a qualitative literature review were carried out. We analyzed 113 contributions published in academic journals since 1990. On the basis of this analysis, we first identified four major areas of research in order to propose a more in-depth understanding of the most relevant issues addressed by the scholars. Then we carried out an expert survey to point out the future research agenda for frugal innovation. In the final section, we draw our conclusions, and we highlight the managerial implications and limitations of the study.

2. Methodology

A Systematic Literature Review (SLR) (Tranfield et al., 2003) is the most suitable literature-based method for pursuing our research aims because it helps to: (i) analyze the progress of a specific stream of research; (ii) evaluate the contribution of a given body of literature to specific topics; (iii) make recommendations for future research; (iv) develop a model or framework; (v) answer specific research questions (Fink, 2005).

We followed the three-stage procedure of planning, execution, and reporting (Fink, 2005; Tranfield et al., 2003). In the planning stage, we defined the specific aims of the research, as well as the key data sources.

With respect to the key data sources, we chose peer-reviewed journals (document type ‘article’) as primary sources, because they can be considered validated knowledge and are likely to have the highest impact in the field. SCOPUS was chosen as database of record, as it is the most comprehensive database of peer-reviewed journals in the fields of management, organization, and social science. Moreover, Google Scholar and Ebsco-Host were used to

identify other possible contributions.

The execution phase of the SLR process consisted of two steps, namely data collection and data analysis. Data collection included the extraction of all materials from the electronic resources. The academic journals were screened in search of articles including at least one of the following keywords in “title, keywords or abstract”: “Frugal innovation”, Frugal, Jugaad, “Jugaad innovation”, “Inclusive innovation”, “Grassroots innovation”, “Resource-constrained innovation”, “Good enough innovation”, “Reverse innovation”, “Low-cost innovation”, “Cost innovation”. The keywords were selected based on both a review of the literature and a brainstorming within the review team.

The bibliographic research was conducted according to the following criteria:

- only articles in the “Social Science and Humanities” category;
- only articles written in English, in order to be able to compare different works;
- only articles published between 1990 and today, because we are interested in the latest studies on the topic;
- only papers and reviews published in academic journals.

The results were then combined with those emerging both from a further snowballing search and from cross-references. The data analysis phase included different activities. First of all, materials were labelled and stored. In this phase, duplications (i.e., journal articles that may appear twice when searching different keywords) were deleted. We read 480 abstracts and we considered only those papers that were relevant to our research. We excluded those articles where the keywords (for example the term “frugal”) were considered in a strictly psychological or sociological sense, i.e. when used with little or no reference to the management domain.

After the screening procedure, 113 contributions were selected (see Annex 1) from 74 different journals mainly belonging to the management field. The fact that journals in other fields turned out to be relevant suggests that the topic of this study spans across different disciplines (Fig. 1).

The next step involved the analysis of each article. In order to provide a structured approach to the review, variables were selected on the basis of the existing literature (Agarwal et al., 2017; Hossain, 2016; Weyrauch and Herstatt, 2016; Brem and Wolfram, 2014), and following the inductive category development approach (Hsieh and Shannon, 2005), whereby researchers immerse themselves in the data to allow new insights to emerge (Kondracki and Wellman, 2002).

This process resulted in the definition of the following categories and variables:

- reference data: year, journal, country (first author's university affiliation);

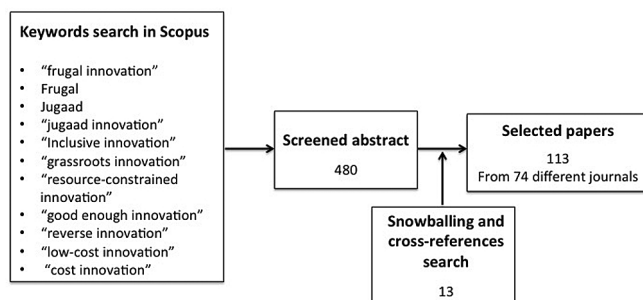


Fig. 1. The articles selection process.

- context issues: countries (emerging and developing, developed), industry;
- methodology issues: types (theoretical vs empirical - single/multiple case study, expert survey, quantitative survey);
- main topic: origins and definitions; ecosystem (institutions and relations among ecosystem actors, innovators at firm-level, innovators at grassroots-level, consumers); innovation process (antecedents/enablers, new product developed (NPD), organizational issues); implementation and diffusion (strategy, impact, application).

Furthermore, data about aim, key findings, and lines of future research were collected and treated as qualitative information to describe the main trends and approaches in this field of study.

“Inter-judge reliability” was measured by the ratio of coding agreements to the total number of coding decisions (Kassarjian, 1977). In this study, three academic judges were involved in the coding process, and the reliability coefficient was 95%.

3. State of the research: quantitative database analysis

As for the year of publication of the selected articles, and considering our time range, our review shows that there are no publications between 1990 and 2004. However, a significant increase in the number of contributions started in 2012 with a peak in 2016. Such data show that this field of research has raised interest in recent years and is likely to raise more in the future (Fig. 2).

An interesting insight emerges from the analysis of the countries where the first author of the articles does his/her research. Indeed, the studies appear to be concentrated in few countries, above all the United Kingdom, followed by India, the United States, and Germany (Fig. 3). The fact that the UK and India are the top two positions may be due to the intense commercial and cultural exchanges that these two countries have traditionally engaged in (many of the authors affiliated to UK Universities come from India). The item “others” includes: Belgium, Denmark, Ireland, Jamaica, Uruguay, Mexico, Philippines, Portugal, Russia, Singapore, Slovakia, South Korea, and The Netherlands.

Empirical studies account for the vast majority of the selected contributions (56%). Among these, almost 74% of the articles adopt multiple (38%) or single (36%) case studies (Fig. 4). The remaining empirical studies are based on quantitative analyses (21%) and on expert surveys (5%). The remaining articles (44%) are theoretical/conceptual papers. Automotive, energy, and telecommunications are the sectors most of the articles focus on. Additionally, the analysis of data reveals that many articles are based on cross-

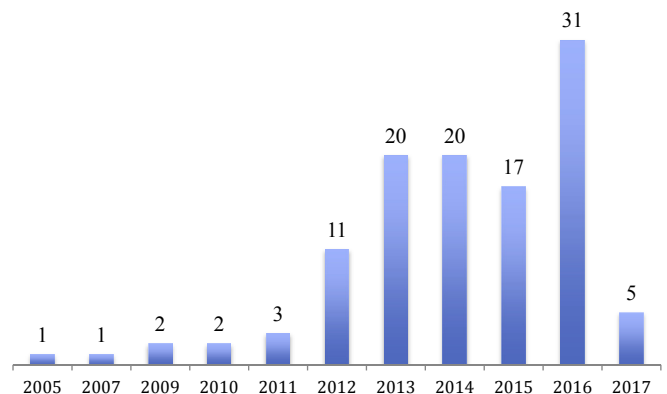


Fig. 2. Number of articles per year. Note: the 2017 data refer to the first quarter of the year.

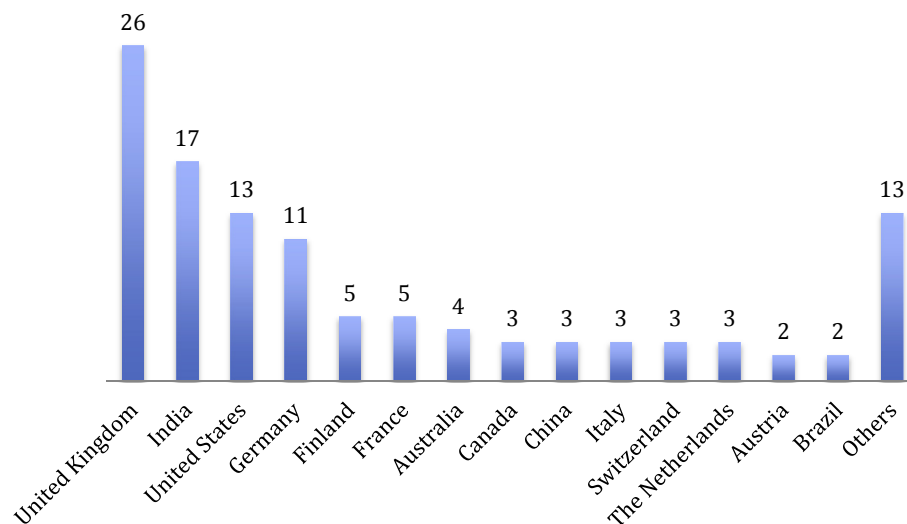


Fig. 3. First author's university affiliation.

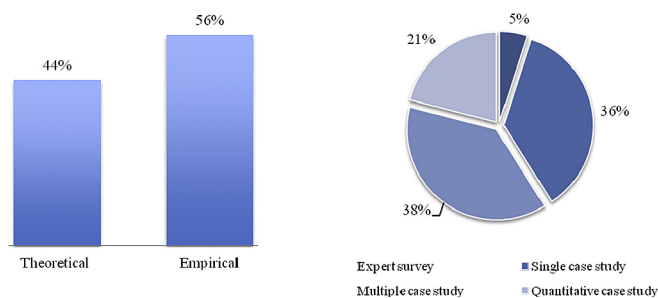


Fig. 4. a. Theoretical and empirical contributions b. Empirical contributions: adopted methodology.

industry samples.

With reference to the geographic focus of the selected articles, we can observe that most scholars concentrate on emerging and developing countries (Fig. 5). Only few articles (5%) compare countries with emerging and developing economies with advanced

ones. A further analysis of the sample shows that 78% of the articles focus on one single country. India ranks first among the countries under investigation (35%), followed by UK (17%), USA (10%), and a number of countries with emerging economies, such as Brazil, China, and Kenya (between 7% and 8%).

4. Qualitative analysis

In this section, we provide a qualitative analysis and a comprehensive discussion on the key findings of the literature. First, we identify four main areas of research: 1) the *origins and definitions of the concept* and the *evolution* of frugal innovation; 2) the *ecosystem* where such innovations emerge and its *main actors*; 3) the *innovation process*; 4) the *implementation and diffusion* of frugal innovations (see Fig. 6). These research areas are analyzed in depth in the following paragraphs.

4.1. Origins and definitions

Frugal innovation has been defined in many ways in the past

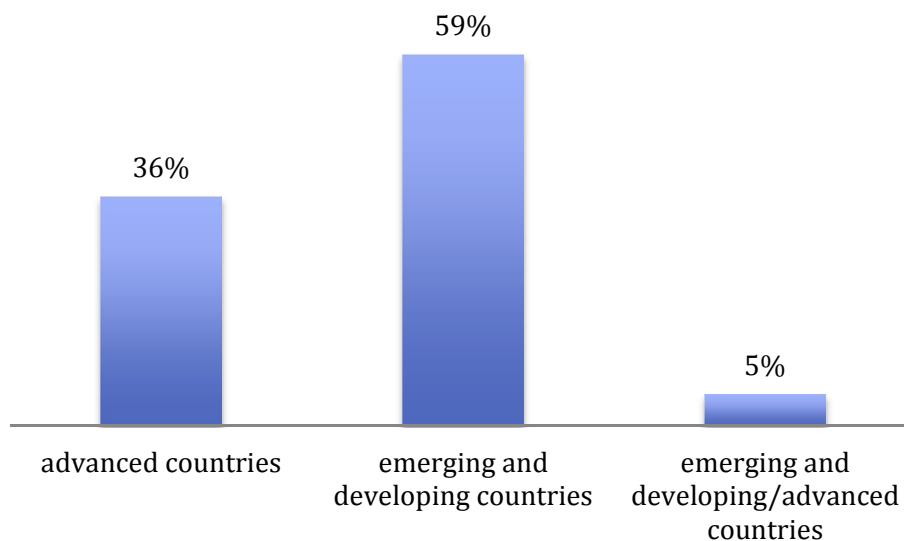


Fig. 5. Geographic focus: advanced vs. Emerging and developing countries.

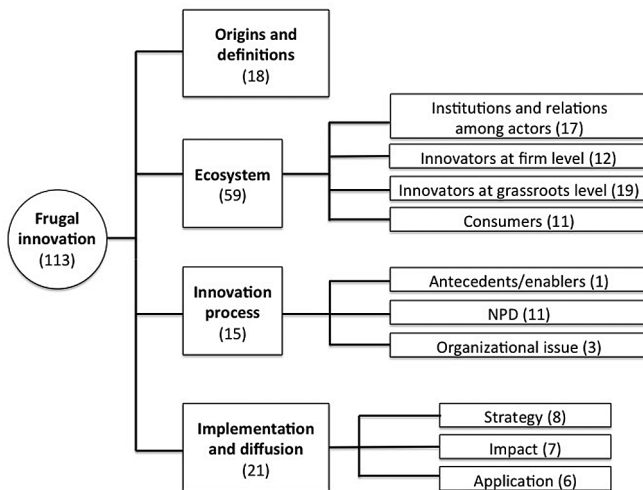


Fig. 6. Main research topics and related articles.

years. Although the various definitions agree on some central features, they show an evolution. Therefore, in order to provide a comprehensive understanding of the term, the definitions existing in the literature have been analyzed over the years.

The origins of the concept of frugal innovation date back to the term “frugal engineering”, coined in 2006 by Carlos Ghosn, Chairman and CEO of the Renault-Nissan Alliance, who succeeded in innovating in a cost-effective and quick way, and under severe resource constraints (Soni and Krishnan, 2014). One of the first definitions was introduced by the Economist in 2010: “Frugal products need to be tough and easy to use (...). Frugal innovation is not just about redesigning products; it involves rethinking entire production processes and business models” (Soni and Krishnan, 2014, p. 31). Afterwards, the topic began to be addressed by scholars, and the SLR has identified three generations of definitions (Fig. 7).

The *first generation* can be described as *product-oriented*, because the definitions point out the product-based features of frugal innovation and highlight the characteristics of frugal products and services. According to these definitions, frugal innovation seeks to minimize the use of material and financial resources, and is characterized by low price, compact design, limited use of raw materials or reuse of existing components, ease of use, and cutting-edge technology to achieve lower costs (Tiwari and Herstatt, 2012; Rao, 2013).

A *second generation* of definitions expands the criteria beyond product features and introduces a set of variables that point out differences and similarities among the various forms of resource-constrained innovations (Zeschky et al., 2014; Brem and Wolfram, 2014; Cunha et al., 2014; Soni and Krishnan, 2014; Agnihotri, 2015). A distinction can be made between *market-oriented* and *process-oriented* definitions. The former underline the fact that frugal innovation is developed in resource-constrained environments to satisfy people at the BoP. More specifically, the research by Zeschky et al. (2014) aimed to understand commonalities and differences among cost, good-enough, frugal, and reverse innovation by using two variables, namely market novelty and technical novelty. They pointed out how frugal innovations originally developed products or services for very specific applications in resource-constrained environments. According to Brem and Wolfram (2014), the different types of innovations can be distinguished by using three variables: sophistication (the attributes reflect a certain object with a specific degree of sophistication and complexity),

sustainability (the attributes involve social and economic responsibility), and emerging market orientation (concerning emerging, developed or international markets). The authors defined frugal innovation as an “approach which focuses on the development, production, and product management of resource-saving products and services for people at the BoP by achieving a sufficient level of taxonomy and avoiding needless cost” (Brem and Wolfram, 2014, p. 19). In this scenario, several scholars have highlighted the dominant role of local contexts, and they have underlined the fact that frugal innovation exists thanks to resource-constrained environments. Agnihotri (2015) identified similarities and differences among frugal, jugaad, value, and reverse innovations, and he concluded that a frugal mindset is needed to develop low-cost innovation. Firms need to “change the mindset of the employees, by changing the culture of the organization” (p. 406). So, according to the author, the culture of a country is the main enabler, and MNCs should be established in low-cost countries. Similarly, Soni and Krishnan (2014) pointed out that “a frugal mindset would be encouraged when an individual or a firm is located in a resource-scarce environment”.

Within the second generation of definitions, other authors suggest a different perspective and provide a *process-oriented* definition whereby frugal innovation is the “means and ends to do more with less for more people” (Radjou and Prabhu, 2015 in Prabhu and Jain, 2015, p. 857) or “the design innovation process that properly considers the needs and context of citizens in the developing world” (Basu et al., 2013).

Finally, the *third generation* represents a breaking point. In fact, Weyrauch and Herstatt (2016) went back to the origin of the concept and identified three criteria that characterize frugal innovation, both in emerging and developed markets: substantial cost reduction, concentration on core functionalities, and optimized performance level. Another important research is that by Agarwal et al. (2017), who identified three fundamental dimensions of constraint-based innovation: cost-effectiveness, ease-of-use, and prescriptive variables.

This return to the origins of the term allows to transfer the concept of frugal innovation from developing countries to advanced economies. Indeed, these authors state that “region-specific definitions and descriptions of these innovations are a limitation of existing literature” (Agarwal et al., 2017, p. 11). A recent study by Rao (2017a) analyzes the use of advances in science and technology to create low-cost sophisticated products. In doing so, he identifies a new class of frugal innovation, namely Advanced Frugal Innovations (AFI). By analyzing 25 examples of AFIs in different sectors and socio-economic contexts, he emphasizes the importance of leveraging advances in science and technology to achieve frugality in resource consumption. These advanced innovations are characterized by three main features: a) low cost, due to the their frugal nature; b) leveraging of advances in science and technology from inception, i.e. involving both product frugal-design and product frugal-engineering; c) involvement of workers with advanced degrees able to understand and leverage advances in science and technology in order to conceive AFIs.

Moreover, the analysis of the criteria that characterize frugal innovation offers “a more profound understanding of the core of frugal innovation, to better meet the requirements of emerging markets and to transfer and adapt the main concept from emerging markets to developed markets” (Weyrauch and Herstatt, 2016, p. 12).

It turns out that frugal is often considered to be an approach (Prabhu and Jain, 2015; Brem and Wolfram, 2014) or a mindset (Soni and Krishnan, 2014) rather than a specific typology of innovation. Therefore, in line with these remarks and with the definition provided by Radjou and Prabhu (2015), we claim that frugal is

First generation Years: 2012/2013	Second generation Years: 2014/2015	Third generation Years: 2016/2017
<i>Product-oriented definitions</i>	<i>Market-oriented definitions</i>	<i>Criteria-oriented definitions</i>
Frugal innovation seeks to minimize the use of material and financial resources in the complete value chain (development, manufacturing, distribution, consumption, and disposal) with the objective of reducing the cost of ownership while fulfilling or even exceeding certain pre-defined criteria of acceptable quality standards (Tiwari and Herstatt, 2012)	Frugal innovations are originally developed products or services for very specific applications in resource-constrained environments. It is often quite disruptive (Zeschky et al., 2014)	Innovations are frugal if they simultaneously meet the criteria substantial cost reduction, concentration on core functionalities, and optimised performance level (Weyrauch and Herstatt, 2016)
Frugal innovation is characterised by: low price, compact design, use of limited raw materials or reuse of existing components, ease of use and use of cutting-edge technology to achieve lower costs (Rao, 2013)	A derived management approach, based on jugaad, which focuses on the development, production, and product management of resource-saving products and services for people at the BoP by achieving a sufficient level of taxonomy and avoiding needless costs (Brem and Wolfram, 2014)	All resource-constrained innovations have the following features: cost-effectiveness, ease-of-use, prescriptive variable (Agarwal et al., 2017)
	Frugal innovation aims to provide the essential functions people seek to satisfy with a given product. For frugal innovators, scarcity is both fact (they cannot easily change the economic condition of their potential customers) and opportunity (Cunha et al., 2014)	Advanced Frugal Innovation (AFI) by leveraging advances in science and technology to capture the frugality inherent in a grassroots frugal innovation (Rao, 2017a)
	Frugal innovation refers to those innovative products and services which are developed under conditions of resource constraints (Agnihotri, 2015)	
	<i>Process-oriented definitions</i>	
	Frugal innovation as the means and ends to do more with less for more people (Radjou and Prabhu, 2015)	
	The design innovation process that properly considers the needs and context of citizens in the developing world (Basu et al., 2013)	

Fig. 7. The evolution of definitions.

to be considered as an approach related to the whole innovation process or product/service characteristics that applies across different typologies of innovation.

A synonym for frugal approach is jugaad, i.e. “the art of overcoming harsh constraints by improvising an effective solution using limited resources” (Radjou et al., 2012 in Rao and Jain, 2015, p. 847). It is commonly considered as an example or a sub-set of frugal innovation that stresses the concept of “improvisation”, and is mainly related to India and China where frugality is one of the key elements along with flexibility and inclusivity (Rao, 2013; Birtchnell, 2013; Holtbrügge, 2013).

Moreover, a frugal approach can be adopted to develop cost innovations, i.e. solutions that provide resource-constrained customers with the same product functionalities but at lower costs (Zeschky et al., 2014). Also, such an approach can develop good-

enough innovations, i.e. solutions that include functionalities and features designed to meet a range of resource constraints beyond capital constraints. Thus, products and services are adapted or re-engineered to fit the specific requirements of low-income markets (Zeschky et al., 2014; Srivastava, 2015). In this perspective, Western firms can develop good-enough products to meet the customer needs of specific growing markets (BoP in advanced markets).

Furthermore, studies have pointed out that frugal is linked to the concept of reverse innovation (Harris et al., 2016; Von Zedtwitz et al., 2015), which refers to the flow of products and services from developing markets to developed markets (Immelt et al., 2009). Thus, innovation is conceived and adopted in emerging economies and it is then ‘exported’ to developed economies (Corsi and Di Minin, 2014). In this regard, scholars have suggested that while

all reverse innovations are also frugal, not all frugal innovations become reverse innovations (Hossain et al., 2016; Rosca et al., 2017).

The frugal approach can also be applied to grassroots innovation – a bottom-up innovation development approach – and it can be combined with inclusive innovation, which is the development and implementation of new ideas aimed at creating opportunities for the enhancement of the social and economic wellbeing of disenfranchised members of society (George et al., 2012).

Finally, bricolage processes – defined as “making do by applying combinations of the resources at hand to new problems and opportunities” (Baker and Nelson, 2005, p. 333) – can be carried out by using a frugal approach.

Some scholars have linked the frugal approach to the concept of disruptive innovation. Among them, Rao (2013) – in his article “How disruptive is frugal?” – aimed to understand why frugal innovation is disruptive. According to the author, the product-based features of frugal innovation put them in a “formidable position to disrupt their incumbents” (Rao, 2013, p. 70). Zeschky et al. (2014) reached the same conclusion and defined frugal innovations as products or services based on new product architectures that are often quite disruptive for entire new customers groups in resource-constrained environments. Empirical evidence of the potential disruptive force of frugal innovations was presented by Ray and Ray (2011) in the context of emerging economies. Following Christensen's work on disruptive innovations, the authors showed that a “frugal use of technology and resources is required for crafting a disruptive technology that provides basic functionalities at a very low price” (Ray and Ray, 2011, p. 224).

Table 1 summarizes the key findings from the literature.

4.2. The ecosystem

Frugal approaches to innovation have been mainly identified in resource-constrained environments, emerging and developing countries, and BoP contexts (see par. 4.1), where the ecosystem plays a crucial role. System of innovation (Sol) frameworks have been identified as useful tools for understanding frugal approaches to innovation (Foster and Heeks, 2013; Arocena et al., 2015). According to the extant literature on Sol, innovation – and, more generally, new knowledge – is expected to emerge from a complex set of elements and interactions – learning, searching, and

exploring processes – among actors within the institutional and social context. With specific reference to the topic under investigation, conventional systems of innovation frameworks have been broadened to incorporate demand-side actors and other actors. Consumers, communities of practice as innovators, local and informal institutions, and innovation intermediaries need, therefore, to be included in the framework (Foster and Heeks, 2013; Arocena et al., 2015).

Against this background, we shall now discuss the main findings arising from the SLR, which focused on the above-mentioned key elements of the ecosystem. We identified the following categories:

- *Institutions and relations among ecosystem actors*: in the first subparagraph, we provide evidence of the role played by “institutions”, considering both the policy-making level, and the role of education and research institutions in the social context;
- *Innovators involved in the frugal innovation process*:
 - ^ *Firm-level innovators*: innovation involving different types of organizations, such as small-, micro-, and medium-sized enterprises, as well as multinational enterprises;
 - ^ *Grassroots-level innovators*: innovation involving individuals, informal enterprises, and local communities.
- *Consumers*: they play a key role as the final beneficiaries of frugal innovations.

Here below, we shall outline the main findings arising from the SLR of each of the above-mentioned elements of the ecosystem.

4.2.1. Institutions and relations among ecosystem actors

Institutions play a critical role in the regulatory framework. As anticipated above, institutions are to be considered with a broader meaning, highlighting the influence of the environment on the actors' behavior. In this respect, a combination of formal regulation and informal customs should be taken into account, especially with regard to BoP/developing countries (Foster and Heeks, 2013). Institutions have been analyzed in terms of formal bodies, formal rules/regulations, but also in terms of informal rules/practices. Such rules do not arise solely from formal institutions; indeed, they exist and change because of the societal influence of the environment where such rules emerge. Formal institutions are closely

Table 1
Origins and definitions – Key findings.

Main topic	Key findings	References
Origins and definitions	- The types of innovations can be distinguished according to different variables: ^ market novelty and technical novelty ^ sophistication, sustainability and emerging market orientation - Differences and similarities among resource-constrained innovations - Frugal innovation could be considered as a mindset, a process, or an outcome - Frugal innovation is characterized by the following criteria: substantial cost reduction, concentration on core functionalities, optimized performance level - The dimensions of constraint-based innovations are: cost-effectiveness, ease-of-use, prescriptive variables - Advanced Frugal Innovation (AFI) highlights frugality in resource consumption during the realization of innovations - AFI leverages advances in science and technologies to capture the frugality inherent in a grassroots frugal innovation	Agarwal et al., 2017; Agnihotri, 2015; Basu et al., 2013; Birtchnell, 2013; Brem and Wolfram, 2014; Corsi and Di Minin, 2014; Cunha et al., 2014; George et al., 2012; Harris et al., 2016; Holtbrügge, 2013; Horn and Brem, 2013; Rao, 2013; 2017a; Roiland, 2016; Soni and Krishnan, 2014; Von Zedtwitz et al., 2015; Weyrauch and Herstatt, 2016; Zeschky et al., 2014 Additional references Baker and Nelson, 2005; Hossain et al., 2016; Immelt et al., 2009; Prabhu and Jain, 2015; Radjou et al., 2012; Ray and Ray, 2011; Rao and Jain, 2015; Rosca et al., 2017; Srivastava, 2015; Tiwari and Herstatt, 2012

intertwined with this complex system of informal institutional elements, which encompasses social, religious, and cultural practices (Pansera and Owen, 2015).

Frugal approaches to innovation cannot be understood without taking into consideration the specificities of the country where such innovations emerge. The unique socio-economic, institutional, and environmental characteristics shape the BoP demand context. Scholars argue that “innovation occurs despite and maybe because of resource constraints” (Pansera and Owen, 2015, p. 309) and “adversity” (Kaur, 2016, p. 320), understood as a combination of poverty, absence of public services, environmental motivations, and social values. In this respect, India is often seen as a potential “ideal laboratory” for conceiving and introducing frugal innovation, not only in resource-constrained contexts (Kaur, 2016; Tiwari and Herstatt, 2012; Shivdas and Chandrasekhar, 2016), but also in the rest of the world. Researchers have underlined the emerging role of India as a “lead market” for frugal innovation (Tiwari and Herstatt, 2012, p. 98). However, the ability of India to reverse its situation, i.e. to become “giver [of innovation] instead of recipient”, is still debated among scholars (Prathap, 2014).

Studies focused on developing countries have linked innovation to concerns on social inclusion (Foster and Heeks, 2013; Sonne, 2012; Baud, 2016), adopting the logic of “from the poor, for the poor” (Pansera and Owen, 2015). Scholars have identified “inclusive and democratizing innovation systems” (Arocena et al., 2015) by considering “knowledge democratization” as one of the means to achieve inclusive development and innovation. Developing countries face considerable difficulties in combining indigenous capabilities/knowledge/innovation with the capabilities to put knowledge at work and achieve satisfactory products and services (Zhang and Mahadevia, 2014; Ustyuzhantseva, 2015). As already mentioned, the literature on innovation systems has stressed how innovation - and knowledge - emerges from the interactions among the actors of the system. Most technological, institutional, and social innovations arise from a combination of knowledge, learning – by doing, by using and by interaction - and interactions among individuals and institutions (Arocena et al., 2015; Foster and Heeks, 2013). In this regard, universities play a critical role in the process of knowledge transfer. They build “cultural capital” by helping young people develop competence, and they support processes of learning, knowledge transfer and innovation to foster inclusive development (Arocena et al., 2015). Another basic element in democratizing innovation systems has been identified in the free flow of information. ICT and the widespread use of mobile phones and the Internet offer an unprecedented knowledge-sharing opportunity for most developing countries (Arocena et al., 2015; Foster and Heeks, 2013; Paunov and Rollo, 2016). Some scholars have studied how companies based in developing countries could benefit from a more widespread adoption/use of the Internet along with policies specifically aimed at supporting the ability of enterprises to use it. This involves a systemic view of innovation, learning and interactivity (Guth, 2005).

Finally, while the SLR highlights how frugal innovation arises from “adversity”, resource-constrained contexts need to face a regulatory and infrastructural gap that hampers the diffusion of innovation (Pansera and Owen, 2015; Govindarajan and Trimble, 2012). This is why scholars still report a misalignment between governmental policies to foster innovation and “desirable” final outcomes (Soni and Krishnan, 2014) (see Table 2).

4.2.2. Firm-level frugal innovators

There are at least three types of organizations when applying a frugal approach to innovation: local/micro companies, small-medium enterprises and multinational enterprises (MNEs). In addition to these, social enterprises have “passion, commitment,

patience and local knowledge” (Prabhu and Jain, 2015, p. 851), but sometimes lack the ability to scale their solutions. Despite the well-known role played by small-medium and local companies in supporting frugal innovation, the literature has mainly focused on how MNEs implement and benefit from the adoption of frugal approaches to innovation in emerging and developing countries (Ray and Ray, 2010).

In this regard, several authors have pointed out that the development of the global economy and the growing importance of emerging markets have prompted an increasingly stronger competition in global markets, forcing changes in MNEs' strategies (Borini et al., 2012; Ray and Ray, 2010). MNEs focusing on low-income markets need to re-think their innovation efforts to meet the “unconventional” demand of these markets (Ojha, 2014, p. 8). Innovating “in” and “for” emerging markets includes co-creating with customers, developing collaborations with local suppliers starting from the design phase (Ray and Ray, 2011), and involving MNE R&D centers in exploring how to innovate for these markets (Jha and Krishnan, 2013). In other words, companies from developed countries need to redesign their R&D governance processes and implement R&D strategies based on frugal innovation principles (Ojha, 2014).

Earlier studies in the field of International Business assumed that innovation originates in developed countries where corporate headquarters are usually located. However, the latest literature has recognized that nowadays MNEs are not looking for subsidiaries limiting their production chain activities to product adaptation and commercialization, but they are rather interested in technological knowledge as well as innovations (Bezerra and Borini, 2015; Borini et al., 2012). By adopting an early subsidiary involvement policy (Baglieri et al., 2014), innovation projects for local markets are more likely to succeed. This is due to the fact that subsidiaries are in the privileged position to rapidly adapt their competences and production processes to the needs of their markets (in terms of quality, cost-effectiveness and time). By doing so, subsidiaries could also introduce new knowledge and practices for the development of new products and processes so as to transfer innovations to the MNE as a whole and not only to local markets (Borini et al., 2012, 2016).

In this respect, subsidiaries are gradually breaking the habit of adopting technologies developed by their parent company, as if they were simply recipients of knowledge. Indeed, they are investing in the development of complementary technological competences (Borini et al., 2012; Li et al., 2013), thus evolving from “local innovators” to “global contributors” (Ojha, 2014).

Several scholars have highlighted the need to leverage technological knowledge from different host locations – in terms of articulation and mobilization of innovation developed by subsidiaries – as a key factor to help MNEs enhancing a sustainable competitive advantage (Bezerra and Borini, 2015; Borini et al., 2012, 2016). By doing so, the potential of reverse innovation – in terms of innovations designed by subsidiaries and ideally absorbed/used by their parent company – means re-thinking the innovation architecture implemented by MNEs in their global structures and strategies (Ray and Ray, 2010; Borini et al., 2012; Ojha, 2014). Knowledge transfer and exploitation is increasingly widespread across global operations within MNEs (Borini et al., 2012, 2016). In this regard, technology transfer follows nontraditional directions, from developing to developed countries and from subsidiaries to parent companies (Borini et al., 2012, 2016). The growing level of technological specialization and skills featured by subsidiaries generate two new flows of innovation: from subsidiaries to parent companies, and from “focal” subsidiaries to other subsidiaries of the multinational group (Baglieri et al., 2014). Furthermore, knowledge transfer/spillovers occur in many different directions as

Table 2

Ecosystem: Institutions and relations among ecosystem actors – Key findings.

Main topic	Key findings	References
Ecosystem – Institutions and relations among ecosystem actors	– Take into consideration the specificities – the institutional and social context – of the country where such innovations emerge – The role of formal and informal institutions – Conventional systems of innovation frameworks need to incorporate demand-side actors – consumers and communities of practice – and other actors, such as localized and informal institutions and innovation intermediaries (information and knowledge brokers) – India as a potential “ideal laboratory” for conceiving and introducing frugal innovation not only in resource-constrained contexts – The role of universities and research institutions in knowledge transfer and “knowledge democratization” to foster inclusive development – The systemic view of innovation: learning and interactivity among the various actors are needed – Free flow of information is crucial. The widespread use of ICT, mobile phones and the Internet offers an unprecedented knowledge-sharing opportunity	Arocena et al., 2015; Baud, 2016; Foster and Heeks, 2013; Govindarajan and Trimble, 2012; Guth, 2005; Hartley, 2014; Kar et al., 2016; Kaur, 2016; Pansera and Owen, 2015; Paunov and Rollo, 2016; Prathap, 2014; Rowthorn et al., 2016; Shivdas and Chandrasekhar, 2016; Sonne, 2012; Tiwari and Herstatt, 2012; Ustyuzhantseva, 2015; Zhang and Mahadevia, 2014 Additional reference Soni and Krishnan, 2014

a dynamic and reciprocal process of knowledge flow between foreign subsidiaries of MNEs and domestic firms (Li et al., 2013) (see Table 3).

4.2.3. Grassroots-level frugal innovators

Grassroots-level frugal innovators are individuals or groups of individuals (communities) who aim at solving a problem through locally available ingenuity and, in so doing, identifying novel solutions. In this perspective, innovation processes arise from the interplay of human ingenuity and frugality (Pansera and Sarkar, 2016). Moreover, scholars who study innovations at this level often refer to community action as a “neglected, but potentially important site of innovative activity” (Seyfang and Smith, 2007, p. 584; Pansera and Sarkar, 2016).

Studies on this topic belong to the field of research related to grassroots innovation – a small but emerging body of literature – that finds increasing application in a variety of contexts. At BoP level, many of these innovators operate under conditions of

resource scarcity, with very limited or no support by formal institutions. Innovations aim to solve rural problems – within small communities – and are easily implemented. The innovation process is based on knowledge, skills and experience that are embedded in communities and individuals outside the formal institutions of research and industry (Jain and Verloop, 2012). In other words, every community has an innate capacity to provide effective solutions to solve the problems it faces every day (Gupta, 2012). Innovators are basically uneducated people, self-employed in the informal sector, and innovation is based on learning by doing and improvising by trials (Gupta, 2012). In resource-constrained contexts, the outputs of grassroots actions are shaped by the process of frugal innovation – i.e. minimizing the use of materials and energy – and products are more socially accepted since they are tailored to the specific requirements/needs/features of their users (Pansera and Sarkar, 2016; Smith et al., 2014).

Research on this topic has been performed mainly by scholars studying the grassroots innovation movement in India, such as the

Table 3

Ecosystem: Firm-level frugal innovators – Key findings.

Main topic	Key findings	References
Ecosystem- Firm –level frugal innovators	– Types of organizations involved in frugal innovation processes: MNEs, SMEs, micro/local enterprises – Headquarters-subsidiary relation and its effect on reverse transfer of innovation: - ˆ strategic orientation of the subsidiary - ˆ strong integration between headquarters and subsidiary, headquarters support; subsidiary autonomy, trust - ˆ entrepreneurial orientation of the corporation - ˆ decentralized development of R&D and innovation - o access to local knowledge – Technology transfer/knowledge spillovers – Influence of the environment on innovation development and transfer – Technological knowledge from different host locations of MNEs: from local innovators to global contributors – MNEs are rethinking their strategies so as to create, transfer and exploit knowledge that is scattered across the company's global operations – MNEs from developed countries: - ˆ how to innovate and make their products more competitive in global settings - ˆ innovating “in” and “for” emerging markets by co-creating with customers and empowering engineers – SMMEs from emerging and developing countries and their contribution to social, environmental, and economic progress within local communities	Bezerra and Borini, 2015; Baglieri et al., 2014; Borini et al., 2012, 2016; Creech et al., 2014; Jha and Krishnan, 2013; Leavy, 2104; Li et al., 2013; Ojha, 2014; Prabhu and Jain, 2015; Ray and Ray, 2010; Sharma and Jha, 2016

one linked to the Honey Bee Network, which provides many case studies of grassroots innovations and practices developed in the country (Gupta, 2012). By providing evidence of the innovative activity in the informal sector in India, they have attempted to institutionalize this type of innovation and integrate it into the formal innovation system through a networking cooperation with formal institutions (Ustyuzhantseva, 2015). This approach also empowers local communities and promotes an inclusive development (Pansera and Sarkar, 2016; Smith et al., 2014). Inclusion/empowerment can happen by helping local communities to do what they already know and to convert their knowledge into innovative frugal products to be sold either directly to the market or to other companies (Gupta, 2012). Moreover, the new knowledge produced in each local context/community can be potentially exploited in other contexts (Smith et al., 2014). However, more studies – both at a national and a cross-country level – are needed to understand the scalability of this type of institutionalization in other local, national and international contexts (Ustyuzhantseva, 2015), as well as its implication/relevance for innovation policies (Smith et al., 2014). Furthermore, frugal innovators at grassroots-level in advanced economies are mainly motivated by environmental and social concerns. Grassroots innovators are looking for eco-innovation – i.e. innovation that provides environmental benefits and reduces energy consumption – and their activity often includes a sense of social justice, social inclusion and community identity. In this respect, the poor are not only those who seek simple and frugal technological solutions to face daily problems in local communities, but they are also a potential reservoir of easy-to-use, affordable and potentially scalable solutions to face environmental/social problems (Pansera and Sarkar, 2016; Seyfang and Smith, 2007).

Frugal innovation at grassroots-level in advanced economies aims not only at reducing production costs and enhancing productivity in a sustainable way, but also at empowering users by leveraging environmental sustainability. Such approaches have been applied in several contexts in advanced economies (Boyer, 2015; Bradbury and Middlemiss, 2015; Hoppe et al., 2015; Kirwan et al., 2013; Monaghan, 2009; Ornetzeder and Rohracher, 2013; Ross et al., 2012; Seyfang and Longhurst, 2013, 2016). In this regard, scholars have identified conceptual frameworks that explain this approach to innovation – specifically in advanced economies – by emphasizing the role of learning within “communities of practice” and explaining how learning (Bradbury and Middlemiss, 2015; Hargreaves et al., 2013; Seyfang et al., 2014) and knowledge dissemination (Seyfang and Longhurst, 2013) occur. In this perspective, the environmental/social nature of frugal approaches to innovation at grassroots level – and the diffusion of practices within networks (Boyer, 2015; Hargreaves et al., 2013; Seyfang et al., 2014) – offers more sustainable/alternative forms of production, consumption and provision (Monaghan, 2009; Bradbury and Middlemiss, 2015; Martin et al., 2015; Ornetzeder and Rohracher, 2013; Ross et al., 2012; Seyfang and Smith, 2007; White and Stirling, 2013).

Scholars demand more research and attention on this approach to innovation. Despite the potential contribution of frugal innovation – developed at grassroots-level – to promoting a sustainable and inclusive growth (Pansera and Sarkar, 2016; Smith et al., 2014), studies in the field have given only a marginal contribution to the mainstream academic debate on sustainability, mainly because there is little evidence of replicability and limited international comparisons of successful practices (Feola and Nunes, 2014) (see Table 4).

4.2.4. Consumers

To study frugal approaches to innovation from a systemic point of view, scholars need to incorporate the demand side. When frugal innovation is approached from the consumer's perspective, two main streams of research can be identified, namely studies focused on BoP consumer behaviors in resource-constrained environments, and studies dealing with the link between consumers' behavior in advanced economies and environmental sustainability issues.

According to the first stream of research, a considerable amount of articles posits that the frugal innovation approach aims at mitigating the lack of resources and finding a new way (value proposition) to fulfill low-income consumers' needs, while ensuring an adequate level of quality (Foster, 2014). When considering the specific BoP context in emerging markets – where poor consumers face a daily struggle for survival in relation to food, energy, housing, transportation, health, ICT and water, – the lack of resources and the unfavorable financial conditions are to be considered as enablers of “Jugaad” (Singh et al., 2012). This approach emphasizes the fact that poor individuals are not passive in their transformation process. Indeed, resource constraints make them act like entrepreneurs, leveraging their skills and indigenous capabilities for innovation (Birtchnell, 2013). In other words, the poor are seen not only as consumers but also as co-producers.

Conversely, the papers belonging to the second stream of research focus on the link between consumers' behaviors and sustainability-related issues, independently from the customers' income. In this perspective, promoting a sustainable development is the main objective of the frugal approach to innovation. This is particularly true for advanced economies. A frugal/sustainable behavior in consumption aims at safeguarding the natural resources and protecting other individuals with altruistic and equitable actions (Bove et al., 2009; Tapia-Fonllem et al., 2013).

Furthermore, scholars studying consumers' behaviors have identified changes in the consumers' approach to innovation and consumption styles where frugality can be seen either as a new emerging lifestyle (Goldsmith et al., 2014) or as a desirable virtue (Bove et al., 2009). In this perspective, the consumers' frugal consumption behaviors could slow down the economic and ecological damage (Bove et al., 2009). Sustainable styles of consumption are found to be influenced by different reasons, such as frugal apparel consumption, fashion consciousness, ecologically and socially conscious consumption (Cho et al., 2015). Similarly, personal networks influence the spending behaviors of frugal consumers, who tend to be more independent than other consumers. However, they are less resistant to social influences when they consume within high-spending personal networks (Lee, 2016).

Recently, researchers have started focusing on the BoP in advanced countries (Angot and Plé, 2015). The purpose is to identify the reasons for not buying in these segments of customers, and to develop affordable, simple and easy-to-use products (Matzler et al., 2014). Frugal innovation makes it possible to tap these markets (Angot and Plé, 2015) (see Table 5).

4.3. The innovation process

The *innovation process* category deals with the innovation process of new products/services. In this research area, the literature mainly focuses on the following topics: 1) innovation enablers/antecedents, 2) characteristics and critical success factors (CSFs) of the new product development (NDP) process, and 3) organizational issues.

As far as the innovation enablers/antecedents are concerned, the literature review highlights the dominant role played by the

Table 4

Ecosystem: Grassroots-level frugal innovators - Key findings.

Main Topic	Key findings	References
Ecosystem- Grassroots-level frugal innovators	- Communities as “niche innovators” - At BoP level: - lack of materials and financial resources - recycling and reuse of pre-existing technologies/frugal assembling of pre-existing technologies with locally produced components - locally embedded knowledge, skills and experience to be leveraged - lack of formal education of innovators and users - learning by doing, by experimenting, trial and errors - low income users - participation of socially excluded groups to the sustainable development “for (and from) the poor” - poor institutional support - potentials of the process of “institutionalization” of grassroots innovation/grassroots innovation in public policy - potential of exploitation, i.e. replication and adaptation of the new knowledge produced in other contexts too - At the level of advanced economies: - environmental motivation/sustainability-driven innovation/sustainable consumption - minimization of complexity in order to guarantee effectiveness and speed in the maintenance of products/processes/systems - clean and affordable innovations to reduce energy consumption - social justice, inclusion and community identity	Boyer, 2015; Bradbury and Middlemiss, 2015; Feola and Nunes, 2014; Gupta, 2012; Hargreaves et al., 2013; Hoppe et al., 2015; Jain and Verloop, 2012; Kirwan et al., 2013; Martin et al., 2015; Monaghan, 2009; Ornetzeder and Rohrer, 2013; Pansera and Sarkar, 2016; Ross et al., 2012; Seyfang and Longhurst, 2013; Seyfang and Longhurst, 2016; Seyfang and Smith, 2007; Seyfang et al., 2014; Smith et al., 2014; White and Stirling, 2013 Additional reference Ustyuzhantseva, 2015

Table 5

Ecosystem: Consumers - Key findings.

Main topic	Key findings	References
Ecosystem- Consumers	- BoP level: - lack of resources and unfavorable financial conditions as enablers of frugal approaches to innovations - low-income consumers' needs and level of quality - simplify over-engineered products to tap growth opportunities in emerging markets - the poor are seen as consumers but also as co-producers - Consumers in advanced economies: - frugal/sustainable consumption to safeguard the natural resources and protect other individuals with altruistic and equitable actions - frugality as a new emerging lifestyle and as a desirable virtue - influence of the personal network on frugal consumption attitudes - frugal apparel consumption, fashion consciousness, ecologically and socially conscious consumption - BoP consumers in advanced countries: detect the reasons for not buying and develop frugal innovation to tap this segment	Angot and Plé, 2015; Birtchnell, 2013; Bove et al., 2009; Cho et al., 2015; Foster, 2014; Goldsmith et al., 2014; Lee, 2016; Matzler et al., 2014; Okulicz-Kozaryn et al., 2015; Singh et al., 2012; Tapia-Fonlleu et al., 2013

ecosystem and the local culture that – in this specific contexts – work as innovation drivers/enablers. Scholars have identified different factors that produce a “resource-constrained” ecosystem and work as enabling factors in the development of affordable products (Pansera and Owen, 2015; Dubey et al., 2015; Soni and Krishnan, 2014; Sharma and Iyer, 2012):

- scarcity of financial and human resources;
- innovation culture;
- inadequate or weak infrastructure;
- weak institutions (or institutional void);
- institutional failure (i.e. patronage relations, female segregation, social exclusion, uneven distribution of resources and unique geography).

By systematically going through the theoretical and empirical literature on NPD, the CSFs and the main contributions to the literature on this topic were identified. The main critical success factors in the process of innovation are summarized below.

- Adoption of the *principles of architectural innovation*, which involves a reconfiguration of the established system so as to link existing components in a new way without changing the core technology or the know-how, but at the same time providing a logical low-cost option for eliminating unnecessary features and creating disruptive technology (Ray and Ray, 2011).
- Implementation of a *modularity approach*. Modularity refers to a core product designed as a platform, where the “architecture”, the “interfaces” and the “standards” remain constant to accommodate future generations of the product (Baldwin and

Clark, 1997; in Ray and Ray, 2011). Modularity provides the innovator with flexibility to customize products to cater to both lower BoP performance requirements, as well as to grow within the more demanding upper-tier mainstream markets (Ray and Ray, 2011).

- Implementation of a *bricolage innovation process* (Linna, 2013). Bricolage can be defined as “applying a combination of the resources whatever at hand to new problems and opportunities” (Baker and Nelson, 2005). Scholars have highlighted that the process of learning comes directly from the field, and the innovation process is not planned but emerges rather spontaneously. The process starts from the problem rather than being consumption-driven (Pansera and Owen, 2015).
- *Leveraging of existing assets* (e.g. networks) (Pansera and Owen, 2015).
- *Exploration of local needs and commitment of managers* to immerse themselves in the local business environment and gain the trust of its actors (Corsi et al., 2014).
- *Development of strategic alliances*. The capabilities acquired in local contexts are critical for success. Companies can access these local capabilities by forging partnerships with local firms and by implementing collaborative partnerships with suppliers and local consultants (Shankar and Hanson, 2013; Lim et al., 2013; Ray and Ray, 2011; Hang et al., 2010). Furthermore, firms can choose to form alliances with mature enterprises that can help companies in the developing country to unfold their intended innovation agenda (Srivastava, 2015). Alternatively, if a firm in a developing country has sufficient financial resources, it can acquire foreign entities with the necessary capabilities to cater to the latent needs of users through augmented products/services (Srivastava, 2015; Lim et al., 2013).
- *Location of the innovation center*. Many scholars agree on the idea that local developers understand local needs better, and many firms shift their innovation hubs from developed countries into emerging markets (Shankar and Hanson, 2013). However, some authors have recently pointed out that successful frugal innovation projects can rely on home-based breakthrough R&D rather than local collaborations or market-based R&D, especially when the transfer of highly technical knowledge is a key factor in the success of the innovation (Altmann and Engberg, 2016).
- In some cases, scholars observe that success can be achieved without financial investments in R&D (Pansera and Owen, 2015), whereas others emphasize the importance – in view of scaling up – of *being prepared to support R&D efforts*. Moreover, the capability to exploit new technologies in new contexts is critical to develop disruptive innovation for emerging contexts. However, since the initial success of a disruptive technology may attract many followers, maintaining innovation requires a robust ongoing R&D effort. Investing in R&D allows the firm not only to develop components or architectures, but also to create relevant knowledge and, as a result, to generate patents (Lim et al., 2013; Hang et al., 2010). Moreover, firms need to invest their resources for pursuing basic and academic research not only in the fields of science and technology, but also in other salient fields, such as business management (Srivastava, 2015).

Another important topic addressed in the literature is that of organizational issues in the innovation process. Key findings in this area point to the importance of team composition and staff motivation (Weiss et al., 2011). Considering diversity within the team can help the innovation process, as innovation in emerging countries is more likely to occur when some managers exhibit familiarity with the local culture (Shankar and Hanson, 2013). Moreover, diversity management can improve the innovation process in terms of increased absorptive capacity, requisite and networking variety,

creative destruction, problem solving and decision-making (Ehrenmann and Warschat, 2013), and a wide range of perspectives (Corsi et al., 2014). Other scholars have emphasized the relevant role of building innovation capabilities (Lim et al., 2013) and knowledge management (Bencsik et al., 2016) (see Table 6).

4.4. Implementation and diffusion

Studies in this category can be grouped in the following main research topics: 1) diffusion strategies and relationship between innovation and business models, 2) impact of the innovation, 3) application of frugal innovation in specific sectors.

Studies on strategies and diffusion are very recent and still fragmented. Some scholars (Hossain et al., 2016) have investigated the different growth strategies by identifying the diffusion patterns of frugal innovation (local, proximity, distance, and reverse). Other authors have specifically addressed reverse strategies and they have demonstrated that reverse innovation can be linked to a premium price strategy (Tournois, 2016). Moreover, scholars have identified two mechanisms behind the MNC decision-making process, namely the perceived degree of needed adaptation and the perceived risk of cannibalization (Zhu et al., 2017). In addition, Agnihotri (2015) explored the potential of the Blue Ocean strategy in emerging markets and concluded that, even though it cannot be stated that blue ocean is better than pursuing a sustainable competitive advantage, this is a useful tool that can help firms to look at new approaches to innovation, such as the frugal one. A recent contribution by Rao (2017b) stressed the importance of the so-called “factor of frugality” to be used in eco-design with the purpose of introducing a change in design philosophy. Due to the focus on reduced use of raw materials, the factor of frugality might also be used as a measure of eco-efficiency.

Another relevant aspect emerging from the literature is the relationship between frugal innovation and business models. Some scholars have highlighted the specificities of frugal and reverse business models and have identified the following features (Rosca et al., 2017):

- target customers range from BOP to middle-class;
- the value proposition focuses mainly on satisfying basic needs;
- the main revenue model is based on low costs;
- the value chain is characterized by cooperation, partnerships and NGO involvement;
- sustainable business models successfully operated at the BoP build upon local resources and capabilities. Thus, the development of local competencies is a success factor.

On this issue, Bocken and Short (2016) investigated how companies might use sufficiency as a driver for innovation. The authors used the concept of frugal innovation to define the sufficiency-driven business model that “delivers sustainability by reducing absolute material throughput and energy consumption associated with provision of goods and services by moderating end-user consumption: encouraging consumers to make do with less” (Bocken and Short, 2016, p. 46).

An emerging topic that has been addressed by many authors in recent years is the social and environmental impact of frugal innovation. More specifically, scholars have highlighted the role and the potential of frugal innovation in promoting environmental sustainability. The following potentials have been identified:

- it encourages to find opportunities beyond the organizations' traditional field of operations and opens new markets (Hyvärinen et al., 2016);

Table 6
Innovation process – Key findings.

Main Topic	Key findings	References
Innovation process: Enablers/Antecedents	Key enablers of innovation: - scarcity of financial and human resources - innovation culture - inadequate or weak infrastructure - institutional failure	Dubey et al., 2015 Additional references Pansera and Owen, 2015; Sharma and Iyer, 2012; Soni and Krishnan, 2014
Innovation process: Critical Success Factors in NPd	Most relevant CSFs: - adoption of an alternative innovation process based on the principles of architectural innovation, modularity approach; - leveraging of existing assets (e.g. networks) - exploration of local needs and commitment of managers - development of a collaborative partnership with local firms, suppliers and local consultants - careful consideration of the location of the innovation center - success can be achieved without financial investments in R&D but investments in R&D are needed for scaling up - support to basic and academic research	Altmann and Engberg, 2016; Bencsik et al., 2016; Corsi et al., 2014; Hang et al., 2010; Lehner and Gausemeier, 2016; Linna, 2013; Ray and Ray, 2011; Shankar and Hanson, 2013; Srivastava, 2015; Williamson, 2010; Xu and Xu, 2016 Additional references Baldwin and Clark, 1997; Baker and Nelson, 2005; Lim et al., 2013; Pansera and Owen, 2015;
Innovation process: Organizational issues	- Properly managing diversity and team climate can provide advantages and business benefits - Framework for building innovation capability - How to apply knowledge management methods	Ehrenmann and Warschat, 2013; Lim et al., 2013; Weiss et al., 2011 Additional references Shankar and Hanson, 2013; Corsi et al., 2014;

- it encourages companies to find new ways of working within their product development and even shift organizational mind-sets and change existing operating structures (Hyvärinen et al., 2016);
- it offers solutions to existing societal problems and its essential outcomes are strictly linked with social inclusion, human well-being, access to essential services, and social coherence (Khan, 2016);
- it can contribute to the establishment of a green supply chain and the development of green products (Sharma and Iyer, 2012);
- in the energy and water sectors, frugal innovations can be seen as more sustainable than existing solutions (Levänen et al., 2016).

Scholars have recently started to measure the degree of sustainability of frugal innovations in the water and energy sectors by developing specific indicators based on sustainable development goals. However, indicators to measure the innovation sustainability should be identified (Levänen et al., 2016; Khan, 2016).

By contrast, some critical positions have emerged from the literature. Some authors have identified the main barriers to developing sustainable frugal innovations. More specifically, the potential advantage of a product of using fewer resources may be partly lost 1) when analyzing the innovations at a larger scale and earlier in their value chain, and 2) because of institutional deficiencies (Hyvärinen et al., 2016). A more critical position is expressed by Rosca et al. (2017). The authors investigated the relationship between frugal, reverse innovation and sustainability, and demonstrated that frugal and reverse innovations do not have an inherent sustainability. As a result, they strongly encourage a “separation of frugal and reverse innovation from sustainability or sustainable business model discussion” (Rosca et al., 2017, p. 12).

As for the social impact, some authors have argued that reverse and frugal innovation can lead to a sustainable, socio-economic change and can contribute to increasing healthy living conditions for the population (Bhattacharyya et al., 2017; Shan and Khan, 2016). However, empirical studies have provided controversial results. They suggest that frugal innovation can have both positive and negative development outcomes (Knorringer et al., 2016).

Finally, a limited number of studies have focused on specific applications of frugal principles to different sectors, with the main goal of finding out benefits and critical factors (Fleming et al., 2016;

Tran and Ravaud, 2016; Barclay, 2014; Gupta et al., 2013; Rao, 2017a; Rai, 2012; Rai, 2015) (see Table 7).

5. Future directions for the frugal approach to innovation research: expert survey

The question now arises about what are the most important future research priorities in the field of frugal innovation. In this respect, we first relate the above-mentioned findings with what we found out to be the major knowledge gaps that represent the need for future research on frugal innovation.

As previously shown, scholars have mainly debated on the factors that drive the adoption of a frugal approach to innovation, but they have also highlighted the gaps and the future steps to be taken in order to spread the implementation of this approach.

For each area of research discussed in the previous sections, we identified the main research gaps suggested by scholars. Then, we developed a survey that was submitted to experts so as to define the most important future research priorities. To do this, we followed a methodology (SLR integrated by expert surveys) that is frequently adopted in various fields of study (Wirtz et al., 2015; Holl et al., 2012; Rabiser et al., 2010; Buchholz et al., 2009). The main goal of our survey was then to validate/understand the importance of each potential future research direction and formulate suggestions for future research in this field (see Table 7).

5.1. Survey design

The questionnaire submitted to experts in the field contains 12 questions, each representing a research gap previously identified through the SLR (Table 8).

Experts were asked to rate the relevance of each of the above items on a five-point Likert scale (from 1 = irrelevant to 5 = essential). Since most scholars have underlined the need to understand whether such approach to innovation could be implemented both in emerging and developing countries and in advanced economies, we asked experts to rate the relevance of the items for both areas.

In the last part of the survey, respondents were asked to indicate additional research directions that might have been overlooked. Thus, we added an open question on other potential topics that could be relevant for future research. Moreover, respondents were

Table 7
Implementation and diffusion - Key findings.

Main Topic	Key findings	References
Implementation and diffusion: strategy and business model	- Identification of patterns of diffusion of frugal innovation (local, proximity, distance and reverse) - Reverse as a premium price strategy - Blue Ocean Strategy useful to look at new approaches - Identification of the factor of frugality to be used in eco-design - Sufficiency as a driver for innovation - Specificities of frugal and reverse business models	Agnihotri, 2015; Bocken and Short, 2016; Colledani et al., 2016; Hossain et al., 2016; Rao, 2017a,b; Rosca et al., 2017; Tournois, 2016; Zhu et al., 2017
Implementation and diffusion: impact	- Potential role of frugal innovations in terms of sustainable development - Indicators to measure the sustainability level of frugal innovation in the water and energy sector - RCPD can enhance the competitive advantage as well as green and supply chain initiatives	Bhattacharyya et al., 2017; Hyvärinen et al., 2016; Khan, 2016; Knorringa et al., 2016; Levänen et al., 2016; Shan and Khan, 2016; Sharma and Iyer, 2012 Additional reference Rosca et al., 2017
Implementation and diffusion: application	- Benefits of the application of frugal principles to different sectors	Fleming et al., 2016; Tran and Ravaud, 2016; Barclay, 2014; Gupta et al., 2013; Rai, 2012; Rai, 2015

Table 8
Expert survey design.

Survey for expert – Items	References
1. Analyze the frugal approach to innovation in SMEs and startups	Hossain, 2016; Brem and Wolfram, 2014; Knorringa et al., 2016; Ray and Ray, 2011
2. Investigate consumer behaviors when approaching frugal products/services	Bove et al., 2009; Cho et al., 2015; Goldsmith et al., 2014; Singh et al., 2012; Tapia-Fonllem et al., 2013; Lee, 2016; Matzler et al., 2014; Agarwal et al., 2017; Hyvärinen et al., 2016
3. Investigate the diffusion and commercialization strategies of frugal products/services	Ray and Ray, 2010; Gupta, 2012; Kirwan et al., 2013; Smith et al., 2014; Goldsmith et al., 2014; Agarwal et al., 2017; Zhu et al., 2017
4. Analyze knowledge management practices to implement the frugal innovation process	Arocena et al., 2015; Pansera and Owen, 2015; Paunov and Rollo, 2016; Kaur, 2016; Baud, 2016; Baglieri et al., 2014; Borini et al., 2012; Li et al., 2013; Rao, 2017a,b
5. How to measure the social impact of frugal innovation	Knorringa et al., 2016; Ehrenmann and Warschat, 2013
6. The contribution of a frugal approach to the implementation of green innovations (product/process/supply chain)	Sharma and Iyer, 2012; Hyvärinen et al., 2016
7. Understanding how to foster collaboration among different actors involved in the frugal innovation process	Arocena et al., 2015; Zhang and Mahadevia, 2014; Boyer, 2015; Seyfang and Smith, 2007; Smith et al., 2014; Pansera and Sarkar, 2016; White and Stirling, 2013; Angot and Plé, 2015
8. Understanding how to measure the environmental sustainability of frugal innovation	Tapia-Fonllem et al., 2013; Khan, 2016; Levänen et al., 2016;
9. Understanding the influence of the cultural context on the diffusion of frugal innovation	Kaur, 2016; Bezerra and Borini, 2015; Smith et al., 2014
10. Better understanding of the political and social barriers to implementing policies that foster frugal innovation	Govindarajan and Trimble, 2012; Knorringa et al., 2016
11. Understanding the role of universities/research centers in supporting companies to implement frugal innovations	Arocena et al., 2015; Creech et al., 2014
12. How to learn from community engagement	Bradbury and Middlemiss, 2015; Feola and Nunes, 2014; Jain and Verloop, 2012; Kirwan et al., 2013; Seyfang et al., 2014; Seyfang and Smith, 2007
Need to understand – for all the aspects under investigation - the contribution of frugal innovation both in emerging and developing countries and in advanced economies	Govindarajan and Trimble, 2012; Cho et al., 2015; Angot and Plé, 2015; Tiwari and Herstatt, 2012; Agarwal et al., 2017; Lim et al., 2013

asked to provide information about their own institution and job position. Since we guaranteed that the responses to the questionnaire would be in anonymous form, the field “name and surname” was optional.

5.2. Survey implementation

We selected 160 experts as key participants in the current debate on frugal innovation and its potential. Experts were identified through the literature review – both academic and practice-oriented – and conference participation lists, and some of them are members of research centers involved in applied research on frugal innovation. They were contacted directly or by e-mail and invited to participate in the survey.

Thirty-six experts answered the survey, with a redemption rate of 22%.

5.3. Survey analysis - results and discussion

As a whole, the results of the expert survey show that both in emerging and in developing countries the role of *SMEs and startups* requires more research (Table 9). The mean value of this item is 4.5, i.e. the highest score in the entire questionnaire. Moreover, it appears that *diffusion and commercialization strategies* (4.36) as well as *how to learn from community engagement* obtained similar mean values (4.33). These are closely followed by the item *foster the collaboration among different actors involved in the frugal innovation process* (4.31).

With regard to the emerging and developing countries, these findings are further supported by the answers given to the open question asking to identify other potentially relevant areas of research on frugal innovation. Indeed, an expert stated: “Grass-root level innovations in emerging economies and how to scale them up”. Another respondent commented: “how convergence between

Table 9

Survey analysis: relevance of each item (mean values).

	Emerging and developing countries (Mean value)	Advanced countries (Mean value)
Frugal approach to innovation in SMEs and startups	4.50	3.97
Consumer behaviors when approaching frugal products/services	4.14	4.11
Diffusion and commercialization strategies of frugal products/services	4.36	4.00
Knowledge management practices to implement the frugal innovation process	4.22	3.92
How to measure the social impact of frugal innovation	4.19	3.69
Contribution of a frugal approach to the implementation of green innovations	4.22	4.33
Foster collaboration among different actors involved in the frugal innovation process	4.31	3.89
Measuring the environmental sustainability of frugal innovation	4.06	4.06
Influence of the cultural context on the diffusion of frugal innovation	4.03	3.86
Political and social barriers to implementing policies that foster frugal innovation	4.06	3.97
The role of universities/research centers	3.64	3.58
How to learn from community engagement	4.33	3.58

developed and developing countries will influence the development and diffusion of (reverse) innovations". In other words, both quantitative and qualitative answers underline that in emerging and in developing countries the most relevant areas of research are related to the diffusion of frugal innovation and to the ecosystem where such an approach to innovation emerges. By specifically considering the ecosystem, the most relevant aspects for future research appear to be the following: firms-level innovators (*SMEs and startups*), the involvement of *communities* in the learning process, and how to *foster the collaboration among the different actors*.

As for advanced countries, *green innovations* (product/process/supply chain) - with a mean value of 4.33 - and *consumer behaviors* - with a mean value of 4.11 - show the greatest significance for future research. These data are also confirmed by suggestions given in response to the open question. In particular, five experts highlighted the importance of better investigating consumer behaviors by suggesting specific topics: "knowledge of customer requirements", "Strategy for consumers' behavioral change from existing industrially mass-produced items to items based on frugality in production", and "location and understanding of consumers who are innovators in consumption of green products". As far as the contribution of a frugal approach to the implementation of green innovations is concerned, one expert emphasized the relevance of "frugal innovation in combination with a circular economy". Third in the ranking is *measuring the environmental sustainability of frugal innovation* (4.06), a topic that was commented by experts in the open question. More specifically, seven experts stressed the importance of measuring the social and environmental impact. A few of them did not provide details, while others suggested measuring specific aspects, such as the "impact of frugality on well-being" or the "mitigation of problems of marginalized groups".

In conclusion, the comparison between the results gathered for emerging and developing countries and those for advanced markets (Fig. 8) showed that the former remain the most relevant context of analysis for future research. For emerging and developing countries, the main topics for future research are the ecosystem and the process of diffusion of frugal innovations. Conversely, for advanced countries, the experts highlighted different topics for future research, namely consumer behaviors and green innovations. Moreover, the respondents underlined another relevant topic, which is related to new product development, and specifically to product features. They suggested to further investigate the "frugal design of product", and to "understand frugality not as a binary concept but as a continuum, comprising different degrees of frugality". This will help the "process for understanding the core elements of innovation prior to replicating them in other contexts".

6. Conclusions

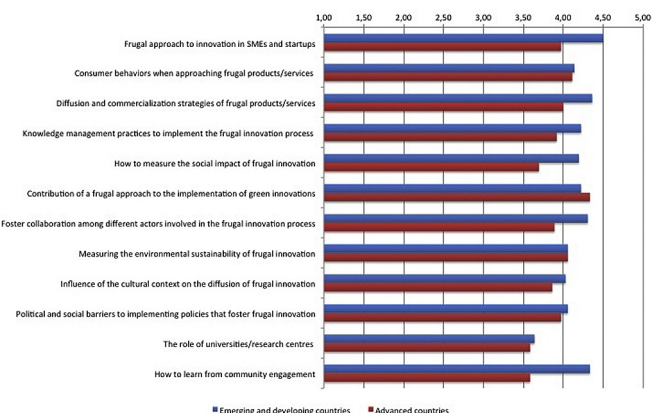
The primary goal of this paper was to organize the extant literature on frugal innovation so as to lay the foundation for subsequent works by identifying gaps in the current knowledge and to recommend new directions for future research. Recent attempts at systematizing the contributions on frugal innovation have confirmed the increasing attention that this topic is drawing among scholars.

To take a step forward and go beyond a mere quantitative review of a selected number of studies on frugal innovation, this paper presented a qualitative analysis of the evolution of the concept and of the key findings that integrate other analyses (Agarwal et al., 2017; Hossain, 2016; Weyrauch and Herstatt, 2016; Brem and Wolfram, 2014) and may help other researchers in developing further empirical work on this issue.

Against this background, we carried out a systematic literature review and analyzed 113 articles published since 1990.

With regard to the quantitative analysis, the results are in line with the findings of previous SLRs on the subject. More specifically, 2012 is confirmed to be the year in which publications on this topic reached a peak after years of slow growth (Agarwal et al., 2017). Moreover, the authors who show the greatest interest in this topic come from the United Kingdom, India, the United States and Germany (Agarwal et al., 2017; Hossain, 2016).

Furthermore, our study confirms that the mainstream literature is focused on emerging and developing countries (Agarwal et al., 2017; Hossain, 2016), and automotive and telecommunications are the most investigated sectors. However, our analysis shows that the energy industry is relevant too.

**Fig. 8.** Survey results: most promising research directions.

The qualitative analysis of the main research areas contributes to the extant literature in different ways. Firstly, in origins and definitions, the *historical evolution of the concept* shows that the authors' heterogeneous perspectives are gradually moving towards a converging understanding of the main characteristics of frugal innovation. In conclusion, we suggest that frugal innovation should be considered as an approach involving the whole innovation process or product/service characteristics rather than a specific typology of innovation. This makes frugal innovation “adaptable” and “transferable” to any industrial and territorial context. The proposed conceptualization is consistent with some common understanding of frugal innovation (Nesta, 2012; 2016; Weyrauch and Herstatt, 2016; Rao, 2017a,b) and supports the latest definitions whereby frugal innovation aims to “create attractive value propositions for their targeted customer groups by focusing on core functionalities and thus minimizing the use of material and financial resources in the complete value chain” (Tiware et al., 2016, p. 17).

Secondly, we provide an advancement in this field of study by giving a literature-based description that can clarify *the relevant dimensions of the ecosystem* where frugal innovation emerges. Frugal approaches to innovation have been mainly identified in resource-constrained environments, in emerging and developing countries, and in BoP contexts. Some scholars (Foster and Heeks, 2013; Arocena et al., 2015) have pointed out that the frameworks of the “conventional system of innovation” need to be broadened to incorporate demand-side actors – i.e. consumers – and other actors, such as communities of practice in their role of innovators, and local informal institutions. Thus, the main research areas we identified and described are institutions and relations among ecosystem actors, frugal innovators – both at firm and grassroots level – and consumers as the final beneficiaries of such innovations.

Thirdly, we identified the crucial role played by the innovation process in defining/framing frugal approaches to innovation. Three sub-topics characterize the *innovation process*: enablers/antecedents of innovation, characteristics and critical success factors (CSFs) of the new product development process, and organizational issues. The literature review focused on new product development pointed out that the critical success factors (CSFs) in the innovation process are mainly related to intangible aspects (such as management of motivations, alliances, etc.).

In addition, we analyzed the extant literature on the *diffusion and implementation* of frugal innovations. As a result of our analysis, the following main issues were identified: the diffusion strategies and the relationship between innovation and business model, the impact of the innovation, and the application of frugal innovation in specific sectors. Hence, the results of the qualitative SLR analysis provide added value to this paper.

Finally, we provided insights into the most promising future

research directions in this field of research. To do so, we asked experts in the field to validate the main research gaps we had identified through the SLR, by rating the relevance of each item both for emerging and developing countries and for advanced economies. The experts' observations contributed to improving our analysis.

As a whole, the results show that experts indicate – for emerging and developing countries and for advanced economies – different relevant future research directions. Specifically, the most promising research area for emerging and developing countries is *the frugal approach to innovation in SMEs and startups*. Moreover, experts demand more research on ecosystem-related aspects, such as *how to foster collaboration among different actors* involved in the frugal innovation process, and specifically on *how to learn from community engagement*, and, last but not least, on the *diffusion and commercialization strategies* of frugal innovations. These results support the findings of the extant literature review, which highlights that research on SMEs is sparse and articles on strategies and diffusion are very recent and fragmented.

As far as advanced countries are concerned, the most promising areas of research appear to be *green innovations* (product/process/supply chain), the *environmental sustainability* of frugal innovation, and the analysis of *consumers' behaviors* when approaching frugal products/services.

Despite its contributions, this paper does have some limitations, above all in relation to the adopted SLR methodology, for example the choice of the keywords and, consequently, the articles included in the analysis, as well as the categorization process (despite the authors' efforts to guarantee a high level of inter-judge reliability). This analysis aims to be illustrative, rather than exhaustive or definitive. Future research providing advancements in the field should include other contributions on the topic, such as book chapters, working papers, and managerial documents/reports.

Nevertheless, the results have important implications both for policymakers and entrepreneurs. In this respect, a better framing of the concept of frugal innovation and the key elements characterizing such innovation allow to identify processes and methodologies that can help companies to better understand how they may benefit from the adoption of a frugal approach to innovation, which can lead to the disruption of entirely new markets. By providing a better value proposition for less affluent customers with a more efficient use of resources, a frugal approach to innovation could generate social and environmental benefits in advanced economies too, even in contexts characterized by austerity.

Annex 1. References of the selected studies for the SLR

N°	Author/s	Year
1	Agarwal, Grottke, Mishra and Brem	2017
2	Agnihotri	2015
3	Agnihotri	2016
4	Altmann and Engberg	2016
5	Angot, Plè	2015
6	Arocena, Göransson and Sutz	2015
7	Baglieri, Bruno, Vasconcellos and Grando	2014
8	Barclay	2014
9	Basu, Banerjee and Sweeney	2013
10	Baud	2016
11	Bencsik, Renáta and Tóth	2016
12	Bezerra and Borini	2015
13	Bhattacharyya, Wu, Mossman, Hayden, Gill, Cheng, Daar, Soman, Synowiec, Taylor, Wong, von Zedtwitz, Zlotkin, Mitchell and McGahan	2017
14	Birtchnell	2011
15	Birtchnell	2013

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N°	Author/s	Year
16	Bocken and Short	2016
17	Borini, Costa and de Miranda Oliveira Junior	2016
18	Borini, de Miranda Oliveira, Silveira and de Oliveira Concer	2012
19	Bove, Nagpal and Dorsett	2009
20	Boyer	2015
21	Bradbury and Middlemiss	2015
22	Brem and Walfram	2014
23	Cho, Gupta and Kim	2015
24	Colledani, Silippo, Yemene, Lanza, Bürgin, Hochdörffer, Georgoulas, Mourtzis, Bitte, Bernard and Belkadi	2016
25	Corsi and Di Minin	2014
26	Corsi, DiMinin and Piccaluga	2014
27	Creech, Paas, Gabriel, Voora, Hybsier and Marquard	2014
28	Cunha, Rego, Oliveira, Rosado and Habib	2014
29	Dubey, Sonwaney, Aital, Venkatesh and Ali	2015
30	Ehrenmann and Warschat	2013
31	Feola and Nunes	2014
32	Fleming, Waweru, Wambua, Ondula and Samuel	2016
33	Foster	2014
34	Foster and Heeks	2013
35	George, Mcgahan and Prabhu	2012
36	Goldsmith, Reinecke Flynn and Clark	2014
37	Govindarajan and Trimble	2012
38	Gupta	2012
39	Gupta, Pandit, Nirjar and Gupta	2013
40	Guth	2005
41	Hang, Chen and Subramian	2010
42	Hargreaves, Hielscher, Seyfang and Smith	2013
43	Harris, Weisberger, Silver, Dadwal and Macinko	2016
44	Hartley	2014
45	Holtbrügge	2013
46	Hoppe, Graf, Warbroek, Lammers and Lepping	2015
47	Horn and Brem	2013
48	Hossain, Simula and Halme	2016
49	Hyvärinen, Keskinen and Varis	2016
50	Jain and Verloop	2012
51	Jha and Krishnan	2013
52	Kar, Padhi and Samantarai	2016
53	Kaur	2016
54	Khan	2016
55	Kirwan, Ilbery, Maye and Carey	2013
56	Knorrtinga, Peša, Leliveld and Van Beers	2016
57	Leavy	2014
58	Lehner and Gausemeier	2016
59	Lee	2016
60	Levänen, Hossain, Lyytinen, Hyvärinen, Numminen and Halme	2016
61	Li, Zhang and Lyles	2013
62	Lim, Han and Ito	2013
63	Linna	2013
64	Martin, Upham and Budd	2015
65	Matzler, Moorandian, Füller and Anschöber	2014
66	Monaghan	2009
67	Ojha	2014
68	Okulicz-Kozaryn, Nash and Tursi	2015
69	Ornetzeder and Rohrer	2013
70	Pansera and Owen	2015
71	Pansera and Sarkar	2016
72	Paunov and Rollo	2016
73	Prabhu and Jain	2015
74	Prathap	2014
75	Rai	2012
76	Rai	2015
77	Rao	2013
78	Rao	2017a
79	Rao	2017b
80	Ray and Ray	2010
81	Ray and Ray	2011
82	Roiland	2016
83	Rosca, Arnold and Bendul	2017
84	Ross, Mitchell and May	2012
85	Rowthorn, Plum and Zervos	2016
86	Seyfang, Hielscher, Hargreaves, Martiskainen and Smith	2014
87	Seyfang and Longhurst	2013
88	Seyfang and Longhurst	2016
89	Seyfang and Smith	2007

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N°	Author/s	Year
90	Shan and Khan	2016
91	Shankar and Hanson	2013
92	Sharma and Iyer	2012
93	Sharma and Jha	2016
94	Shivdas and Chandrasekhar	2016
95	Singh, Gupta and Mondal	2012
96	Smith, Fressoli and Thomas	2014
97	Soni and Krishnan	2014
98	Sonne	2012
99	Srivastava	2015
100	Tapia-Fonllem, Corral-Verdugo, Fraijo-Sing and Durón-Ramos	2013
101	Tiwari and Herstatt	2013
102	Tournois	2016
103	Tran and Ravaut	2016
104	Ustyuzhantseva	2015
105	Von Zedtwitz, Corsi, Søberg and Frega	2015
106	Weiss, Hoegl and Gibbert	2011
107	Weyrauch and Herstatt	2016
108	White and Stirling	2013
109	Williamson	2010
110	Xu and Xu	2016
111	Zeschky, Winterhalter and Gassmann	2014
112	Zhang and Mahadevia	2014
113	Zhu, Zou and Xu	2017

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